UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 004 · Particular integral Type 2 (sin ax, cos ax) · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S004·9	Unit 2 deck 004 ("Particular integral of Type 2"), slide 9 (rendered .pptx)
T004 00:10:32	Unit 2 AV-summary transcript "Particular integral of standard functions – Type 2" (24.7 min), segment start
T004b 00:08:13	Unit 2 AV-summary transcript "Particular integral Type 1 and Type 2" (12.6 min, newer recording); its Type 1 part is tagged T004a in the class 003 pack
CW U2 P13	Unit 2 class-work problem 13
ESA Dec-23 3a	Past paper: UE22MA141A Dec 2023, question 3.a
ESA Jan-24 s1 2d	Past paper: UE23MA141A Jan 2024 set 1, question 2.d

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1 The Type 2 rule: replace D^2 by $-a^2$

Definitions

Type 2 right-hand side. $X=\sin(ax+b)$ or $X=\cos(ax+b)$, with constants a and b. S004·3 · S004·4 · T004 00:00:00

$f(-a^2)$. The expression obtained from $f(D)$ by replacing only the D^2 terms by $-a^2$ (minus a squared). D^3 is written $D^2\cdot D$ and D^4 as $(D^2)^2$ first. T004 00:01:15 · T004 00:01:53 · T004b 00:06:32

Rationalising (conjugate). If a single D remains, as in $\frac{1}{p+qD}$, multiply numerator and denominator by $p-qD$ to get $\frac{p-qD}{p^2-q^2D^2}$; then replace D^2 by $-a^2$ again and let $p-qD$ act on X (differentiate). S004·9 · T004 00:11:48 · T004b 00:06:59

Statements in the class material

> *"Type (ii); When* $X = \sin(ax+b)$ *or* $\cos(ax+b)$*:* $\frac{\sin(ax+b)}{f(D^2)} = \frac{\sin(ax+b)}{f(-a^2)}$ *provided* $f(-a^2) \neq 0$*;* $\frac{\cos(ax+b)}{f(D^2)} = \frac{\cos(ax+b)}{f(-a^2)}$ *provided* $f(-a^2) \neq 0$*."* S004·4

Key points

1. Replace only D^2, never D. Write $D^3 = D^2 \cdot D$ and replace the D^2 part. ESA ★★★ T004 00:01:15 · T004 00:01:53 · T004b 00:06:32
2. "It is not minus 2 the whole square. It is minus into 2 square, which is minus 4." ESA ★★★ T004 00:11:33 · T004b 00:06:32
3. If a D remains after the replacement, rationalise with the conjugate so that D^2 appears, then replace again. ESA ★★★ S004·9 · T004 00:11:48 · T004b 00:06:59
4. $\frac{1}{1+4D}$, $(1+4D)$ are operators: the function always stays to the right of the operator. ESA ★★ T004 00:12:15
5. After rationalising, the numerator operator $(p-qD)$ differentiates X: $(1+4D)\cos(2x+3) = \cos(2x+3) - 8\sin(2x+3)$. ESA ★★★ S004·9 · T004 00:13:23
6. The PI has the same frequency as X; a sine input can give a mixture of sine and cosine (a phase shift), Figure G1.1. ESA ★★ ADDED FOR CLARITY S004·9

Why?

Why may D^2 be replaced by $-a^2$?
$D^2 \sin(ax+b) = -a^2 \sin(ax+b)$, and the same for cosine, so on these functions D^2 acts as multiplication by $-a^2$. Hence $f(D^2)\frac{\sin(ax+b)}{f(-a^2)} = \sin(ax+b)$. A single D turns sine into cosine, so it cannot be replaced by a number.

ADDED FOR CLARITY S004·4

Why multiply by the conjugate?
$(p+qD)(p-qD) = p^2 - q^2D^2$ contains only D^2, which can be replaced by $-a^2$, a number. The conjugate moves to the numerator, where it is just differentiation.

S004·9 · T004 00:11:48 · T004b 00:11:29

Exam traps

Exam trap. $-a^2$ is not $(-a)^2$. For $a=2$, $D^2 \to -4$, so $D^3 + 1 \to 1 - 4D$ (slide 004·9). T004 00:11:33 · S004·9

Exam trap. Do not replace a lone D by a or by ia: rationalise it (E1.3). ADDED FOR CLARITY T004b 00:06:59

2 Case of failure: $f(-a^2) = 0$

Definitions

Case of failure. $f(-a^2) = 0$; it happens when $D^2 + a^2$ is a factor of $f(D)$. S004·5 · T004 00:02:42

$f'(D)$. The derivative of $f(D)$ with respect to D, as if D were a variable: $(D^2 + 3)' = 2D$. T004 00:03:26 · S004·16

Resonance. Forcing at a frequency already present in the CF; the PI carries an extra factor x and its amplitude grows without bound. ADDED FOR CLARITY S004·5

Statements in the class material

> *"Case of failure; If $f(-a^2) = 0$, then $\frac{\sin(ax+b)}{f(D^2)} = x.\frac{\sin(ax+b)}{f'(-a^2)}$ provided $f'(-a^2) \neq 0$; $\frac{\cos(ax+b)}{f(D^2)} = x.\frac{\cos(ax+b)}{f'(-a^2)}$ provided $f'(-a^2) \neq 0$."* S004·5

> *"If $f'(-a^2) = 0$, then $\frac{\sin(ax+b)}{f(D^2)} = x^2.\frac{\sin(ax+b)}{f''(-a^2)}$ provided $f''(-a^2) \neq 0$ (and the same for $\cos(ax+b)$); "similarly, it proceeds on when $f''(a) = 0$....."."* S004·6 · S004·15

Key points

1. $f(-a^2) = 0$ exactly when $D^2 + a^2$ is a factor of $f(D)$, i.e. when $\pm ai$ are AE roots. Look at the CF first: it tells you in advance which terms will fail. ESA ★★★ T004 00:02:53 · S004·16
2. On failure, write x outside the operator and differentiate the denominator with respect to D: $\frac{\cos\sqrt{3}x}{D^2+3} = x\frac{\cos\sqrt{3}x}{2D}$. ESA ★★★ S004·16 · T004 00:23:34 · T004b 00:08:45
3. $\frac{1}{D}$ means integrate, with no constant: $\frac{x}{2D}\cos\sqrt{3}x = \frac{x}{2}\int \cos\sqrt{3}x\,dx = \frac{x\sin\sqrt{3}x}{2\sqrt{3}}$. ESA ★★★ S004·16 · T004 00:22:41 · T004b 00:09:05
4. The x stays outside: only the trigonometric function is integrated, not x times it. ESA ★★ ADDED FOR CLARITY S004·14 · T004 00:23:44
5. When only one factor of $f(D)$ fails, apply the other factor first (E3.5), then the failing one. ESA ★★ S004·13 · T004 00:21:50
6. ESA $(D-2)^2y = 8(e^{2x} + \sin 2x)$: e^{2x} fails twice (Type 1), $\sin 2x$ does not fail (Type 2 with $D^2 = -4$ leaves $-4D$). ESA ★★★ ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a

Why?

Why does an x appear when $f(-a^2) = 0$?
Then $\sin ax$, $\cos ax$ solve the homogeneous equation (they are in the CF), so no multiple of them can give X. Multiplying by x gives a new function; the rule's $x\frac{1}{f'}$ is the limit of $\frac{1}{f}$ as the frequency approaches the root (same idea as Type 1).
ADDED FOR CLARITY S004·5 · T004 00:02:53

Exam traps

Exam trap. Slide 004·15 writes $f'(a)$ in the failure formula; it means $f'(D)$ with $D^2 \to -a^2$ (slides 004·5–6), never f' at the number a. ADDED FOR CLARITY S004·15 · S004·5

Exam trap. For $\frac{1}{D^2+a^2}\sin ax$ the answer is $-\frac{x\cos ax}{2a}$: the integral of $\sin$ brings a minus sign. ADDED FOR CLARITY T004 00:23:44

3 Products, powers and sums: reducing to Type 2

Definitions

Transformation formula. A product-to-sum identity such as $\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$. S004·7 · T004 00:05:42

Superposition. $\frac{1}{f(D)}(X_1+X_2) = \frac{1}{f(D)}X_1 + \frac{1}{f(D)}X_2$, so PI = PI_1+PI_2. S004·11 · T004 00:17:28

Statements in the class material

> *"The same formula can be used to find the PI for the following functions also. When* $X = \sin ax.\cos bx$ *or* $\cos ax.\cos bx$ *or* $\sin ax.\sin bx$*; When* $X = \sin^2 ax$ *or* $\cos^2 ax$*; When* $X = \sin^3 ax$ *or* $\cos^3 ax$*."* S004·7

Key points

1. No formula exists for $\cos^2\frac{x}{2}$ itself: rewrite it with an identity first, then use the rule. ESA ★★★ T004 00:20:16 · T004 00:20:36
2. The constant from $\sin^2$ or $\cos^2$ is $\frac{1}{2}e^{0x}$: Type 1 with $D \to 0$, and it fails if D is a factor of $f(D)$ ($\frac{1}{2(D^3-2D^2+D)} \to \frac{x}{2}$). ESA ★★★ S004·13 · S004·17 · T004 00:21:33 · T004b 00:09:42
3. Split a mixed right-hand side into PI_1, PI_2, ... and find each separately; the PI is their sum. ESA ★★★ S004·11 · T004 00:17:28
4. If the question says "find the PI", stop at the PI; if it says "solve", write y = CF + PI. ESA ★★ T004 00:24:12

Exam traps

Exam trap. Slide 004·14 (and the lecture) give the constant part of PI_2 for $(D-1)^2(D^2+1)y = \cos^2\frac{x}{2}$ as $\frac{1}{4}$; it is $\frac{1}{2}\cdot\frac{1}{f(0)} = \frac{1}{2}$. ADDED FOR CLARITY S004·14 · T004 00:21:33

Exam trap. Slide 004·17's boxed answer drops the 2 of $\frac{\cos 2x}{2(\dots)}$: the correct PI is $\frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}$ (as in CW P18), not $\frac{x}{2} - \frac{8\cos 2x}{100} + \frac{6\sin 2x}{100}$. ADDED FOR CLARITY S004·17 · T004b 00:10:35 · CW U2 P18

Exam trap. $\sin^2 x = \frac{1-\cos 2x}{2}$ (frequency doubles), not $\frac{1-\cos x}{2}$ as said at T004b 00:09:26. ADDED FOR CLARITY S004·17 · T004b 00:09:26

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences; 7 distinct / 9 occurrences are PI by operator rules). Type 2 sits in 5 distinct questions (7 occurrences): $(D-2)^2y = 8(e^{2x} + \sin 2x)$ in Dec-23 3a = May-24 3b = Jun-24 2a (★★★, 6–7 marks, solved in full here, W2.3), and the Type 2 term of the mixed questions Jul-23 3b ($\sin x$), Jan-24 set 1 2d ($\cos 2x$ with a failure), set 3 2c ($\sin^2 x$, = CW P18 = slide 004·17) and set 4 2c ($\sin^2 \frac{x}{2}$), whose other terms are solved in the class 005 and 006 packs (table C3.2). Stars: the rule, rationalising and the failure case ★★★; $\sin^2/\cos^2$ ★★★ (three questions); products $\sin A \cos B$ and $\sin^3$ ★ (slides only, never asked); the second failure (x^2, f'') ★ (never asked).

1 · The Type 2 rule: replace D^2 by $-a^2$

ESA Jul 2023 · 7 marks · UE22MA141A Q3.b · solved in W1.3

Solve $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$. Type 2 part: $\frac{\sin x + \cos x}{10}$. Full solution: class 005 pack.

2 · Case of failure: $f(-a^2) = 0$

ESA Dec 2023 · 7 marks · UE22MA141A Q3.a (= May 2024 Q3.b, June 2024 Q2.a) · solved in W2.3

Solve $(D-2)^2y = 8(e^{2x} + \sin 2x)$, where $D = \frac{d}{dx}$.

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q2.d · solved in W2.2

Solve $(D^2 + 4)y = x^2 + \cos 2x + 2^{-x}$. Type 2 part (failure): $\frac{x \sin 2x}{4}$. Full solution: class 005 pack.

3 · Products, powers and sums: reducing to Type 2

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q2.c (= CW P18) · solved in W3.2

Solve $(D^3 - 2D^2 + D)y = x^2e^{-x} + \sin^2 x$. $\sin^2 x$ part: $\frac{x}{2} - \frac{\cos 2x}{25} + \frac{3 \sin 2x}{100}$. Full solution: class 006 pack.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q2.c (= CW P22) · solved in W3.5

Solve $[(D-1)^2(D+1)^2]y = \sin^2 \frac{x}{2} + 2^{-x} + x$. $\sin^2 \frac{x}{2}$ part: $\frac{1}{2} - \frac{\cos x}{8}$. Full solution: class 005 pack.